

Technology Stack

Date	30 June 2026
Team ID	*****
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Smart Lender identifies the programming languages, frameworks, storage, front-end tools, back-end tools, and supporting libraries used to build an AI-powered loan approval prediction system. The chosen stack ensures the application is scalable, maintainable, and aligned with the project's machine learning requirements.

The table below details the specific technologies chosen for each component of the architecture, along with the justification for each choice.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript (with Flask Jinja templates)	A lightweight HTML/CSS/JS form provides a simple, responsive interface for credit officers to enter applicant details and instantly view the prediction without the overhead of a separate frontend framework.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python with Flask	Flask integrates seamlessly with the Python-based ML stack (scikit-learn, XGBoost), allowing the trained model to be loaded and served for real-time predictions through simple REST routes.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV dataset for training; joblib-serialized model and scaler files for storage	Applicant training data is structured and tabular, so a CSV file is sufficient for model training; the trained XGBoost model and scaler are persisted as .pkl files for fast loading during inference, avoiding the overhead of a full database for this use case.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	IBM Cloud	IBM Cloud offers straightforward deployment for Flask-based ML applications with scalable compute resources, making it cost-effective for hosting the Smart Lender prediction service.

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5	Version Control & CI/CD (e.g., <i>GitHub, GitLab, Jenkins</i>)	GitHub	GitHub enables seamless collaboration between the team (Yaswanth Ganesh R, Sandhya Mondam, Salma Shaik) for tracking code changes, managing branches for model experiments, and coordinating contributions.
6	Third-Party APIs / Other Tools (e.g., <i>Stripe, Twilio, Google Maps</i>)	NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, SciPy	These Python libraries enable data preprocessing, model training (Decision Tree, Random Forest, KNN, XGBoost), evaluation, and visualization, forming the core machine learning pipeline behind the creditworthiness predictions.